Problem 7.1: A horizontal pipeline has a diameter of 0.1 m. Water is pumped through this pipeline with a mean velocity V=1.35 m/sec. Kinematic viscosity index of water $\nu=10^{-7}$ m^2/sec. Roughness of the boundary a=1.5 10^{-4} m.

a) What difference in head can be read from two manometers that are connected to the pipe and of which the distance between the points of connection equals to 100 m.

b) Same question for a liquid with a viscosity index. $\nu=10^{-5}$ m^2/s

c) Same question for a liquid with a viscosity index. $\nu=10^{-4}$ m^2/s

1. Akım türbülant mı, laminar mı? ($Re = VD/\nu$)
   - $Re < 2320$: Laminar ($V = 0.5 u_{max}$)
   - $Re > 2320$: Turbulent ($V = 0.81 u_{max}$)

2. for turbulent; thickness, $Re_\delta = \frac{u_* \delta_a}{\nu} = 11.6 \Rightarrow \delta_a = 11.6\nu/u_*$

Viscous Sublayer	Overlap Layer	Outer Layer
$\tau = \mu \frac{du}{dy}$	$\tau = \mu \frac{du}{dy} - \rho\overline{u'v'}$	$\tau = -\rho\overline{u'v'} = \rho\varepsilon\frac{du}{dy}$

3.

Hydraulically Smooth	Transition	Hydraulically Rough
$k < \delta_a$	$\delta_a < k < 6\delta_a$	$k > 6\delta_a$
→ C depends on: ν	→ C depends on: ν + k	→ C depends on: k
$\rightarrow \frac{V}{u_*} = 1.75 + 5.75 \log \frac{u_* D}{2\nu}$	$\rightarrow \frac{V}{u_*} = 5.75 \log \frac{12R}{k + 0.35\delta_a}$	$\rightarrow \frac{V}{u_*} = 4.75 + 5.75 \log \frac{D}{2k}$
hepsi için pezeti		

a) $\nu = 10^{-6}$ m^2/s , $Re = \frac{VD}{\nu} = \frac{1.35 \times 0.1}{10^{-6}} = 1.35 \times 10^5 > 2320$: Turbulent ⇒ Chezy kullanılır

! Hydraulically rough kabulü, Δh bulmak için $\Delta h'$i bilmemiz gerektiğinden sorunun kalanını bu varsayımla götürüp en son teşit edeceğiz. ($R = D/4 = 0.025$ m)

$k > 6\delta_a$, $V = C\sqrt{RJ}$ — chezy

$u_* = \sqrt{gRJ} = 3.13\sqrt{RJ}$, $k = 2a$, $\frac{6R}{a + \delta_a/7}$

$\frac{V}{u_*} = 5.75 \log\left(\frac{12R}{k + 0.35\delta_a}\right) \Rightarrow V = \underbrace{(3.13\sqrt{RJ}) \times 5.75}_{\approx 18\sqrt{RJ}} \log\left(\frac{12R}{k + 0.35\delta_a}\right) = C\sqrt{RJ}$

$\Rightarrow C = 18 \log \frac{6R}{a} = 18 \log \frac{6 \times 0.1/4}{1.5 \times 10^{-4}} = 54\ m^{1/2}/s$, $V = C\sqrt{RJ}$, $\Delta h = JL$; $\Delta h = \frac{LV^2}{C^2 R} = \frac{100 \times 1.35^2}{54^2 \times 0.025} = 2.5$m

! başta yaptığımız kabulü kontrol et, $k > 6\delta_a$

$J = \Delta h / L = 2.5/100 = 0.025$

$u_* = \sqrt{gRJ} = \sqrt{10 \times 0.025 \times 0.025} = 0.079$ m/s

$\delta_a = 11.6\nu/u_* = 11.6 \frac{10^{-6}}{0.079} = 1.47 \times 10^{-5}$m

$k = 3 \times 10^{-4}$, $6\delta_a = 8.8 \times 10^{-5}$; $k > 6\delta_a$ ✓ kabul doğru

b) $\nu = 10^{-5}$ m^2/s : $Re = VD/\nu = \frac{1.35 \times 0.1}{10^{-5}}$, $1.35 \times 10^4 > 2320$: Turbulent

ezâne bulduğumuz değeri kayıp deniyoruz ; $\Delta h = 2.5$m , $u_* = 0.079$ m/s

$\delta_a = 11.6 \frac{10^{-5}}{0.079} = 1.47 \times 10^{-3} \rightarrow 6\delta_a = 8.8 \times 10^{-3} > k = 3 \times 10^{-4}$ ⇒ya transition'da ya da hydraulically smooth

! burada, verdiğimiz Δh değerine kabul görecek kadar yakınlaşana dek iterasyon yapılır.

1.iterasyon : $\Delta h = 3$m (uymayan Δh değerinden biraz fazla bir değer denenir)

$J = \frac{\Delta h}{L} = \frac{3}{100} = 0.03$

$u_* = \sqrt{gRJ} = \sqrt{10 \times 0.025 \times 0.03} = 0.087$ m/s

$\delta_a = 11.6\nu/u_* = 11.6 \times 10^{-5}/0.087 = 1.34 \times 10^{-3}$m

$C = 18 \log \frac{6R}{a + \delta_a/7} = 18 \log \frac{6 \times 0.025}{1.5 \times 10^{-4} + 1.34 \times 10^{-3}/7} = 47.6$

$\Delta h = \frac{LV^2}{C^2 R} = \frac{100 \times 1.35^2}{47.6^2 \times 0.025} = 3.22$m

$3.22 \neq 3.00 \Rightarrow \Delta h_2 = 3.22$ (2. iterasyon)

2.iterasyon : $\Delta h = 3.22$m

$J = \frac{\Delta h}{L} = \frac{3.22}{100} = 0.0322$

$u_* = \sqrt{gRJ} = \sqrt{10 \times 0.025 \times 0.0322} = 0.09$ m/s

$\delta_a = 11.6\nu/u_* = 11.6 \times 10^{-5}/0.09 = 1.29 \times 10^{-3}$m

$C = 18 \log \frac{6R}{a + \delta_a/7} = 18 \log \frac{6 \times 0.025}{1.5 \times 10^{-4} + 1.29 \times 10^{-3}/7} = 47.3$

$\Delta h = \frac{LV^2}{C^2 R} = \frac{100 \times 1.35^2}{47.3^2 \times 0.025} = 3.20$m

$3.20 \simeq 3.22 \Rightarrow \Delta h = 3.2$m kabul edilir

$k = 3 \times 10^{-4} < \delta_a = 1.3 \times 10^{-4}$: Hydraulically Smooth

c) $\nu = 10^{-4}$ m^2/s : $Re = VD/\nu = \frac{1.35 \times 0.1}{10^{-4}} = 1350 < 2320$: Laminar

Laminar flow → Hagen Poiseuille (Chezy türbülant için)

$V = \frac{gI}{\nu} \frac{D^2}{32}$, $\Delta h_L = \frac{64}{Re} \frac{L}{D} \frac{V^2}{2g} = 4.32$m

($\lambda = \frac{64}{Re}$) → Darcy-Weisbach Eq.

ν (m^2/s)	Re	Regime	Δh over 100 m
10^{-7}	1.35×10^6	Turbulent, hydraulically rough	2.5 m
10^{-5}	1.35×10^4	Turbulent, hydraulically smooth	3.2 m
10^{-4}	1350	Laminar	4.32 m

Problem 7.2: In a welded steel oil line (diameter 0.6m), pumps are spaced 10 km apart. The pumps are working a difference in pressure head of 25 m oil column. Calculate the discharge. Density of the oil is $\rho=800$ kg/m^3, kinematic viscosity is $\nu=10^{-4}$ m^2/s, boundary roughness is $a=5\times10^{-5}$ m. In the course of time, the roughness of the pipe (expressed in terms of roughness length will grow 3×10^2 times the original value. Will the discharge be influenced by this? If so, how much (in %) will the reduction (approximate) be?

$D = 0.6$m ; $L = 10^4$m ; $\Delta h = 25$m ; $\rho = 800$ kg/m^3 ; $\nu = 10^{-4}$ m^2/s ; $a = k/2 = 5 \times 10^{-5}$m ; $a' = 300a$

1. $J = \frac{\Delta h}{L} = \frac{25}{10^4} = 2.5 \times 10^{-3}$, $R = D/4 = 0.15$m

! bu noktada, turbulent flow kabulü yaparak soruyu çözeceğiz, en son bu kabulün doğrulağunu kontrol edeceğiz.

2. $u_* = \sqrt{gRJ} = \sqrt{10 \times 0.15 \times 2.5 \times 10^{-3}} = 0.061$ m/s

$\delta_a = 11.6\nu/u_* = 11.6 \frac{10^{-4}}{0.061} \simeq 0.02$m

$k = 10^{-4}$ m $<<$ δ_a ($= 0.02$m) ⇒ hydraulically smooth (ν'ye bağlı. (a veya k yok)

3. $C = 18 \log \frac{6R}{\delta_a/7} = 18 \log \frac{6 \times 0.15}{0.02/7} = 45\ m^{1/2}/s$

4. $V = C\sqrt{RJ} = 45\sqrt{0.15 \times 2.5 \times 10^{-3}} = 0.87$ m/s

$Q_1 = V \frac{\pi D^2}{4} \simeq 0.246\ m^3/s$

$Re = VD/\nu = \frac{0.87 \times 0.6}{10^{-4}} = 5200 > 2320$: Turbulent ✓

~ old pipe : $a' = 300a = 15 \times 10^{-3}$ m ; $k' = 2a' = 3 \times 10^{-2}$m

Δh, J ve δ_a aynı , şimdi transition'da ⇒ $\delta_a = 0.02 < k' = 0.03 < 6\delta_a = 0.12$m

1. Transition : C hesabında, a ve δ_a kullanılır.

$C = 18 \log \frac{6R}{a' + \delta_a/7} = 18 \log \frac{6 \times 0.15}{15 \times 10^{-3} + 0.02/7} = 30.6\ m^{1/2}/s$

2. $V' = C\sqrt{RJ} = 30.6\sqrt{0.15 \times 2.5 \times 10^{-3}} = 0.59$ m/s

$Q' = V' \frac{\pi D^2}{4} = 0.167\ m^3/s$

$\frac{0.246 - 0.167}{0.246} \times 100 = 32\%$ reduction